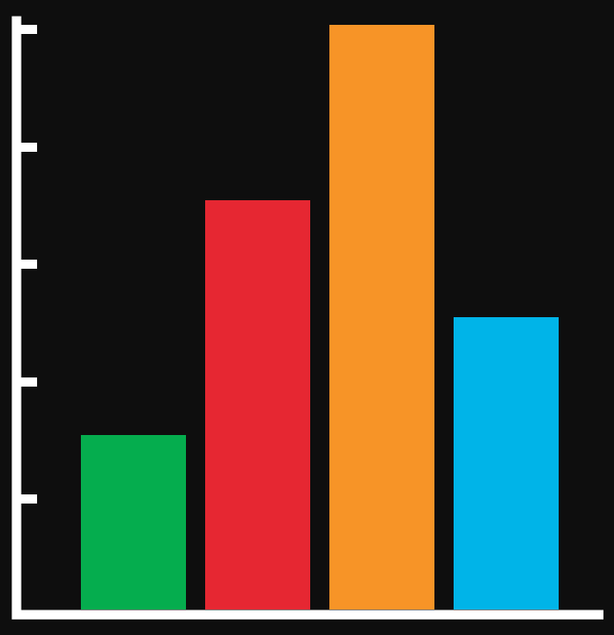
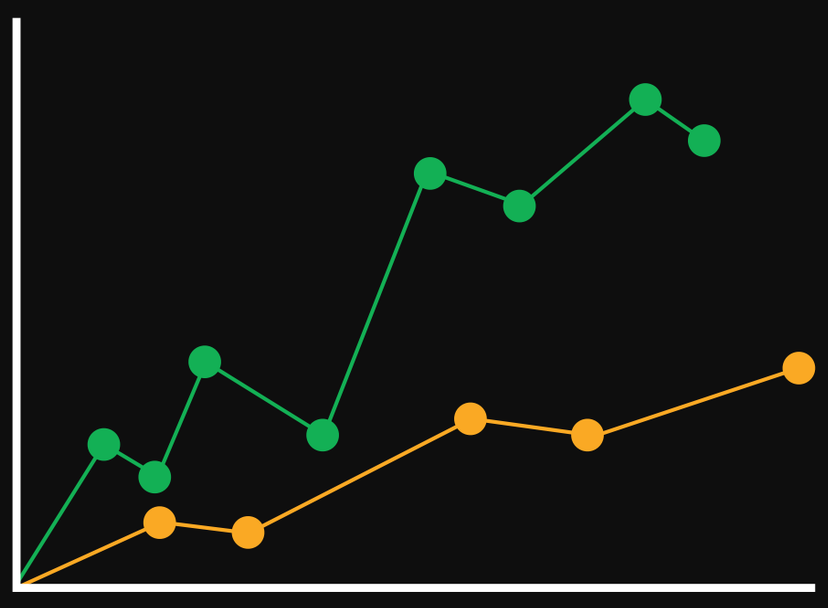
WHICH CHART WHEN?

The Data Analyst's Guide to Choosing the Right Charts

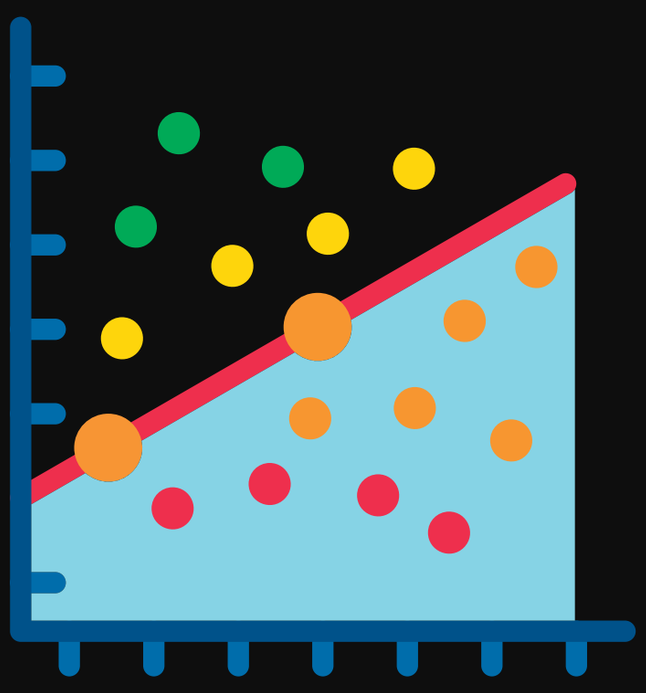


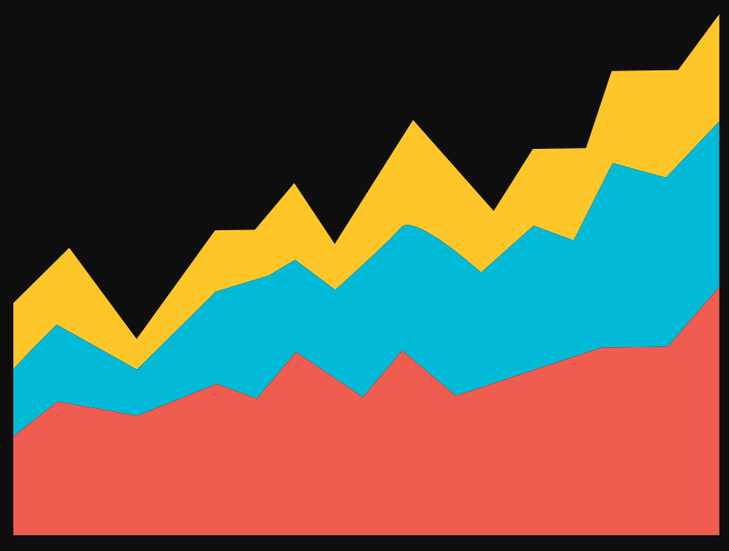
ANDREW MADSON

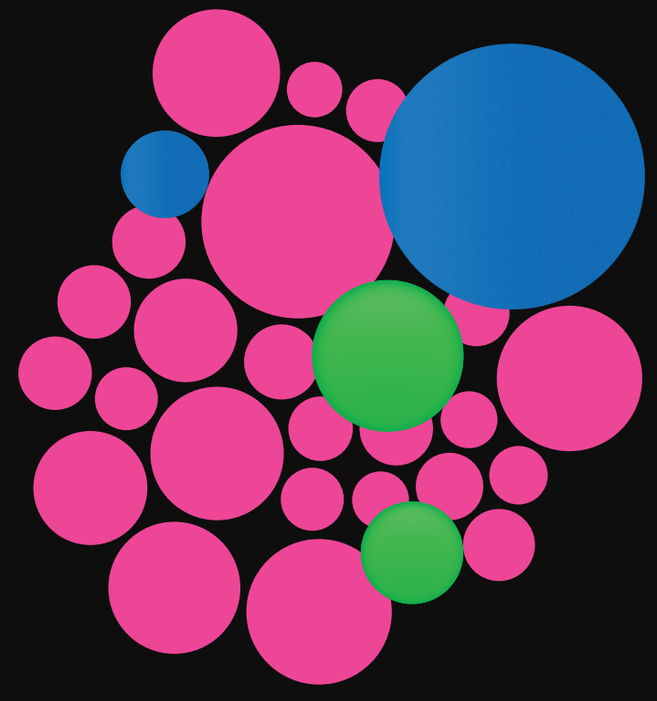


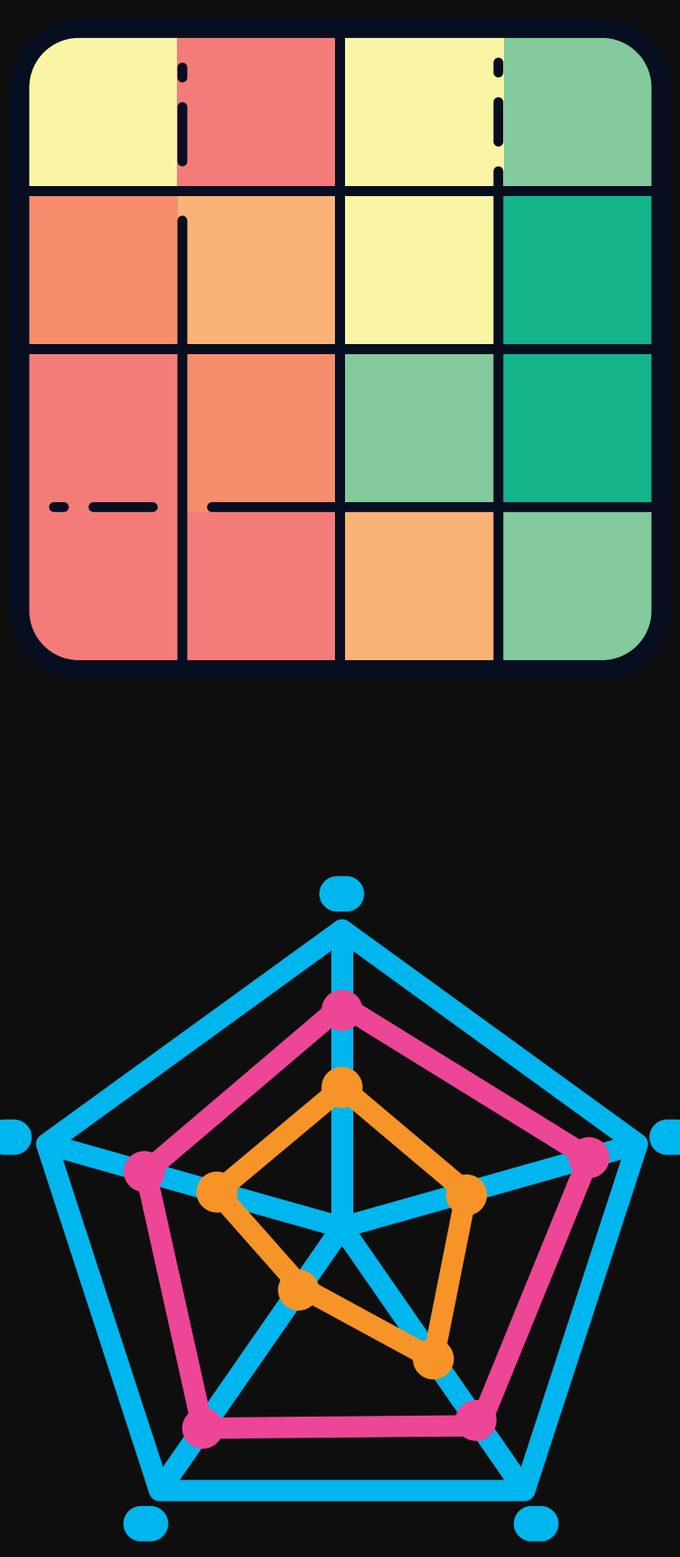


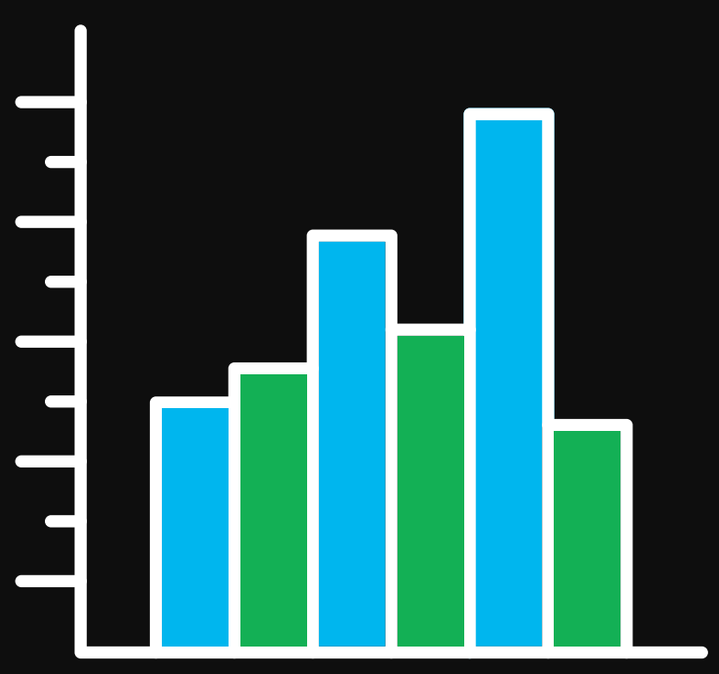


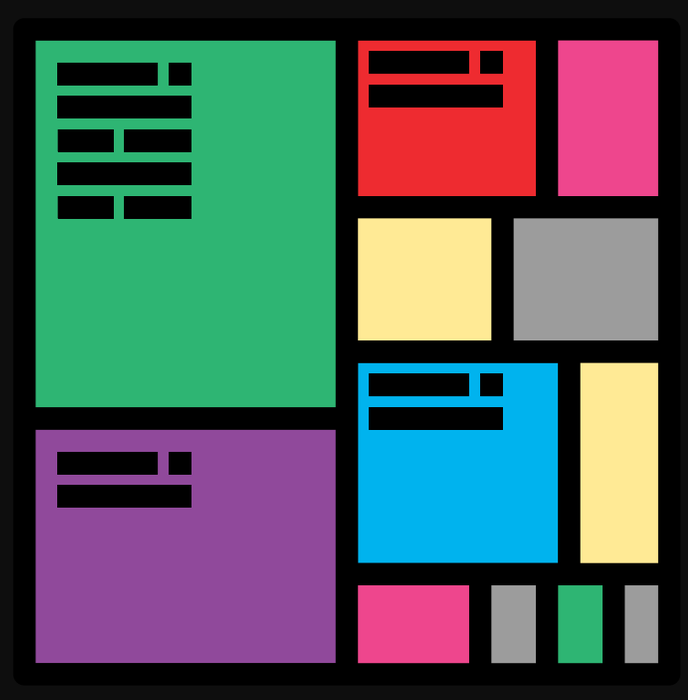


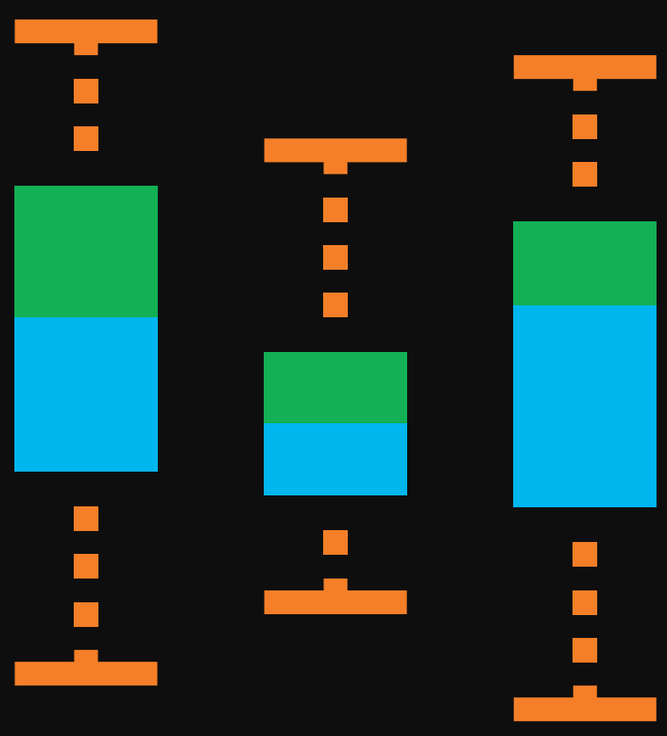


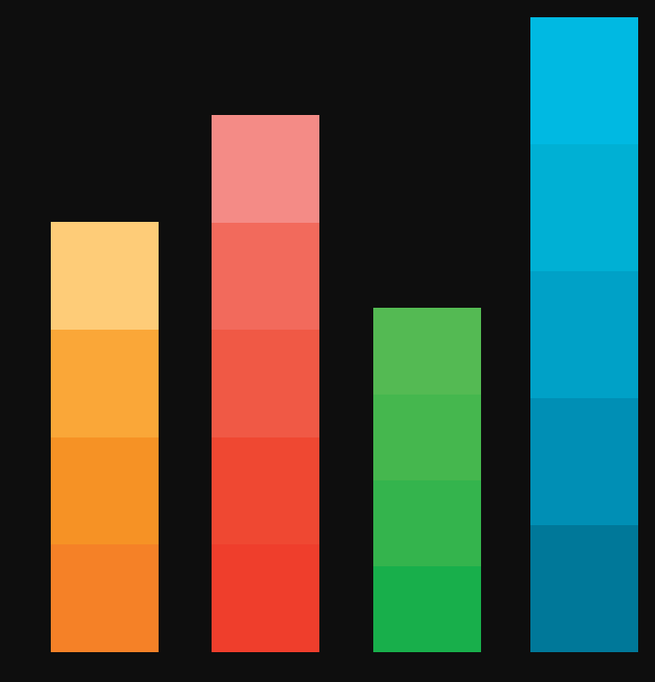


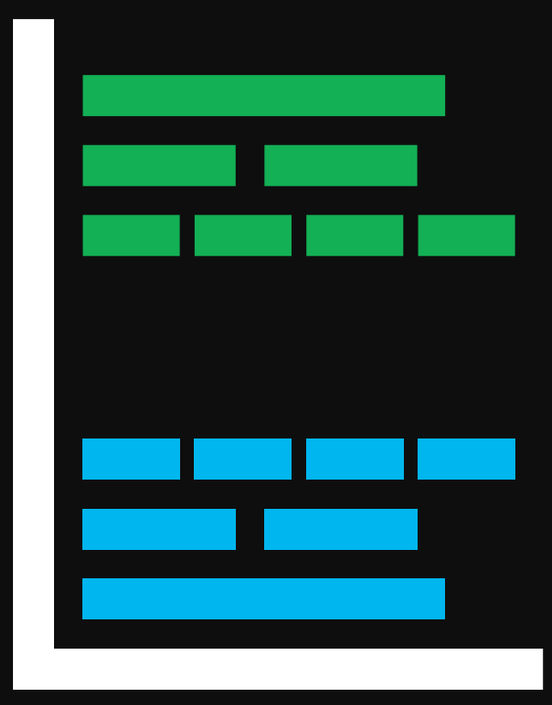


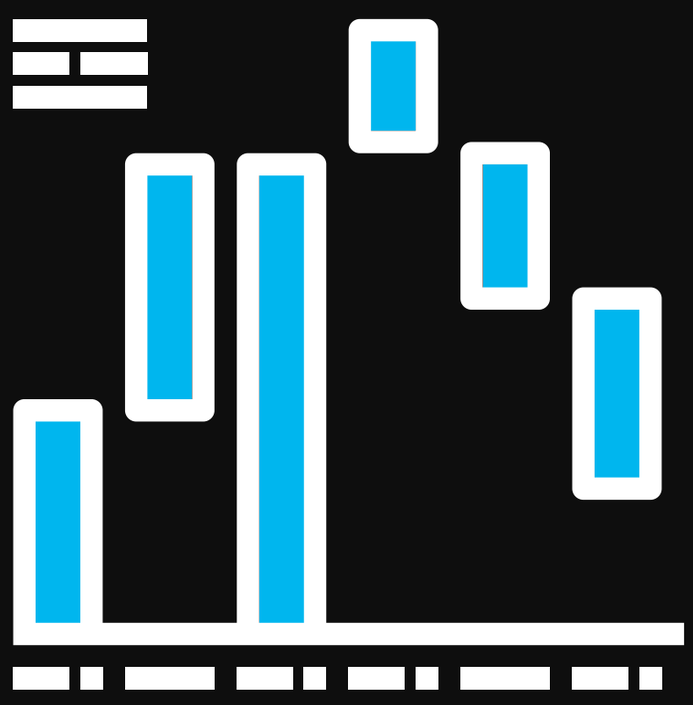


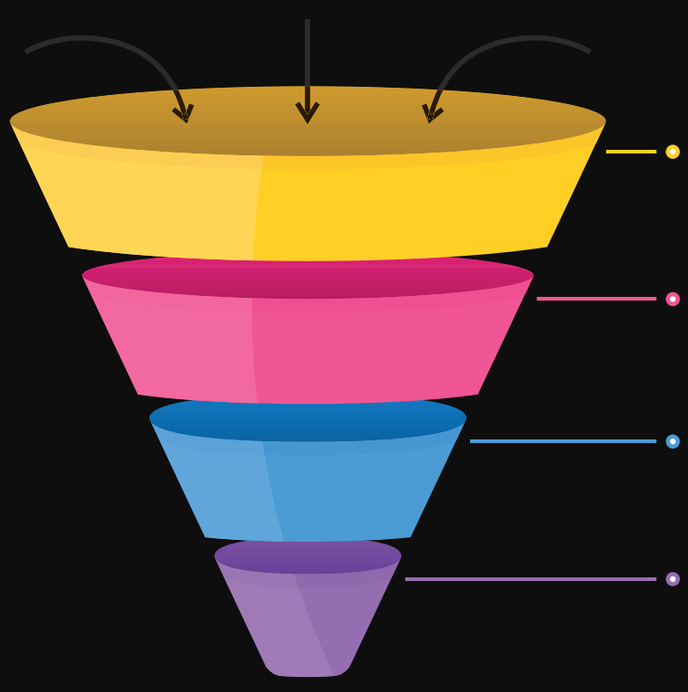












Introduction: Unlock the Power of Your Data!

Are you tired of spending hours crafting the perfect data visualization only to have your audience squinting, scratching their heads, or simply losing interest? As a data analyst, you know how crucial it is to present your findings in an easy-to-understand and engaging manner. After all, the insights you uncover are only as powerful as your ability to communicate them effectively.

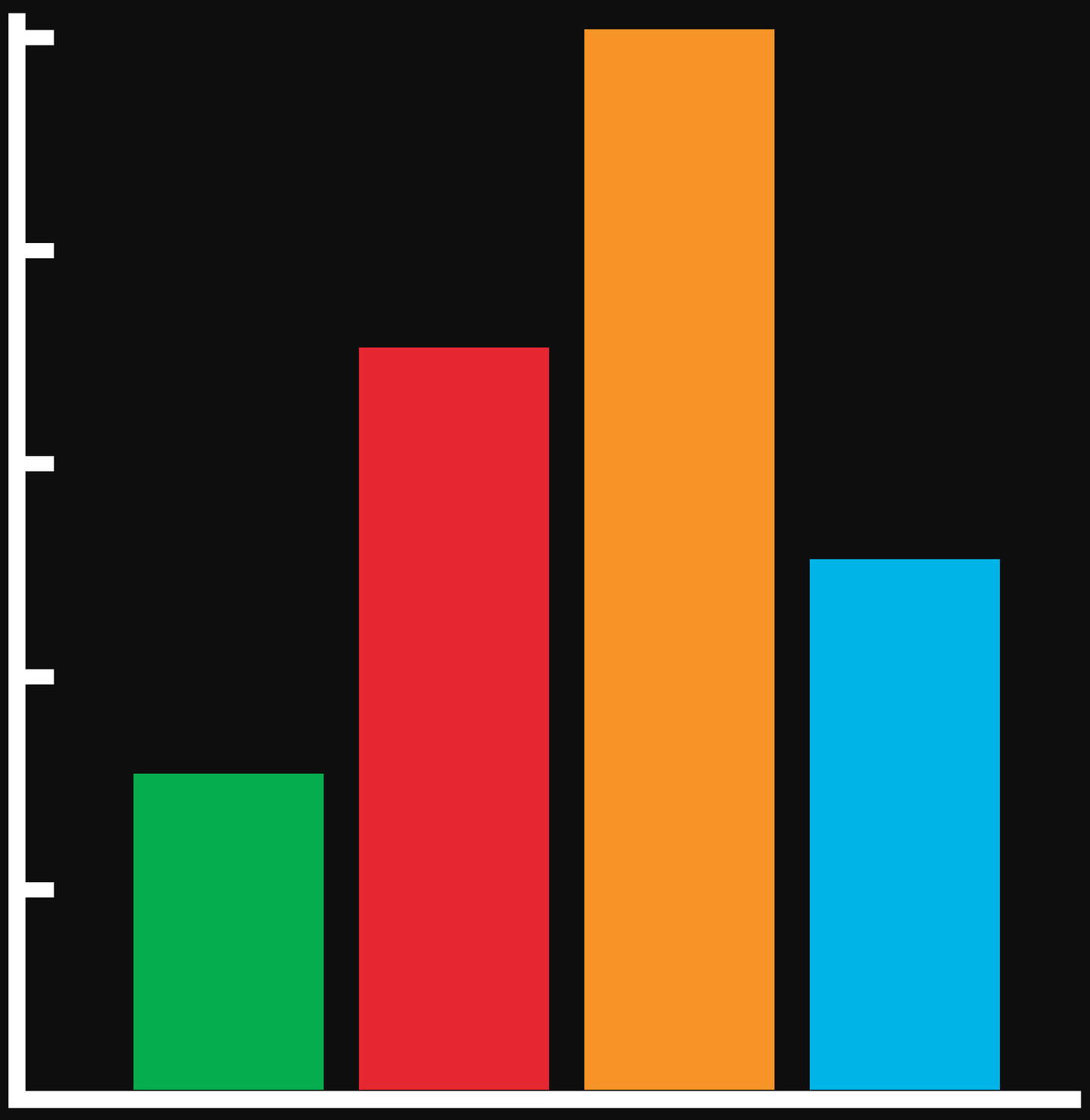
Fret not, fellow data enthusiasts! I've got you covered. This guide is your ultimate resource for choosing the right charts to bring your data to life. No more confusion or glazed-over eyes—just clear, compelling visualizations that will have your audience eagerly following along and enjoying the journey through your data.

By reading this guide, you'll discover the following:

- How to identify the perfect chart type for your specific data and goals
- 15 popular chart types, explained in detail, with tips on when and how to use each one
- How to avoid common pitfalls that can make your visualizations less effective

Keep your valuable insights from getting lost in translation. Dive into this guide and unlock the full potential of your data by mastering the art of choosing the right charts. Your audience will thank you, and your career as a data analyst will soar to new heights!

1. Bar Chart



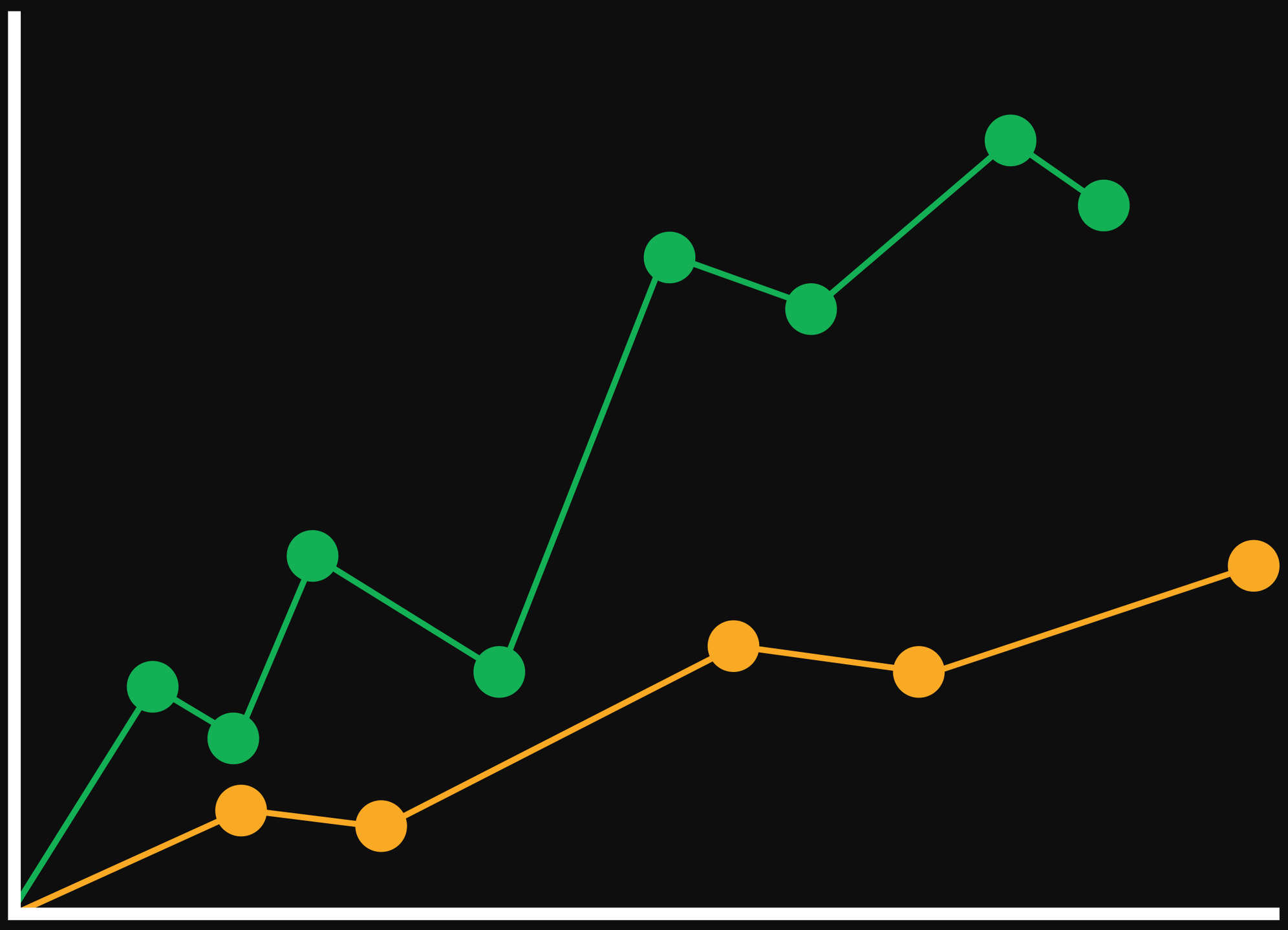
- **Type of data:** Categorical, quantitative

- **When to use it:** Use a bar chart to compare data across categories.

- **What it shows:** Bar charts display data using rectangular bars, with the length of the bar representing the value. The bars can be horizontal or vertical.

- **When to avoid it:** Avoid using a bar chart when there are too many categories or if the data is continuous.

2. Line Chart



- **Type of data:** Continuous, time-series

- **When to use it:** Use a line chart to show trends over time.

- **What it shows:** Line charts plot data points connected by lines. The X-axis usually represents time, and the Y-axis represents the value.

- **When to avoid it:** Only use a line chart when there is a logical order or relationship between data points.

3. Donut Chart



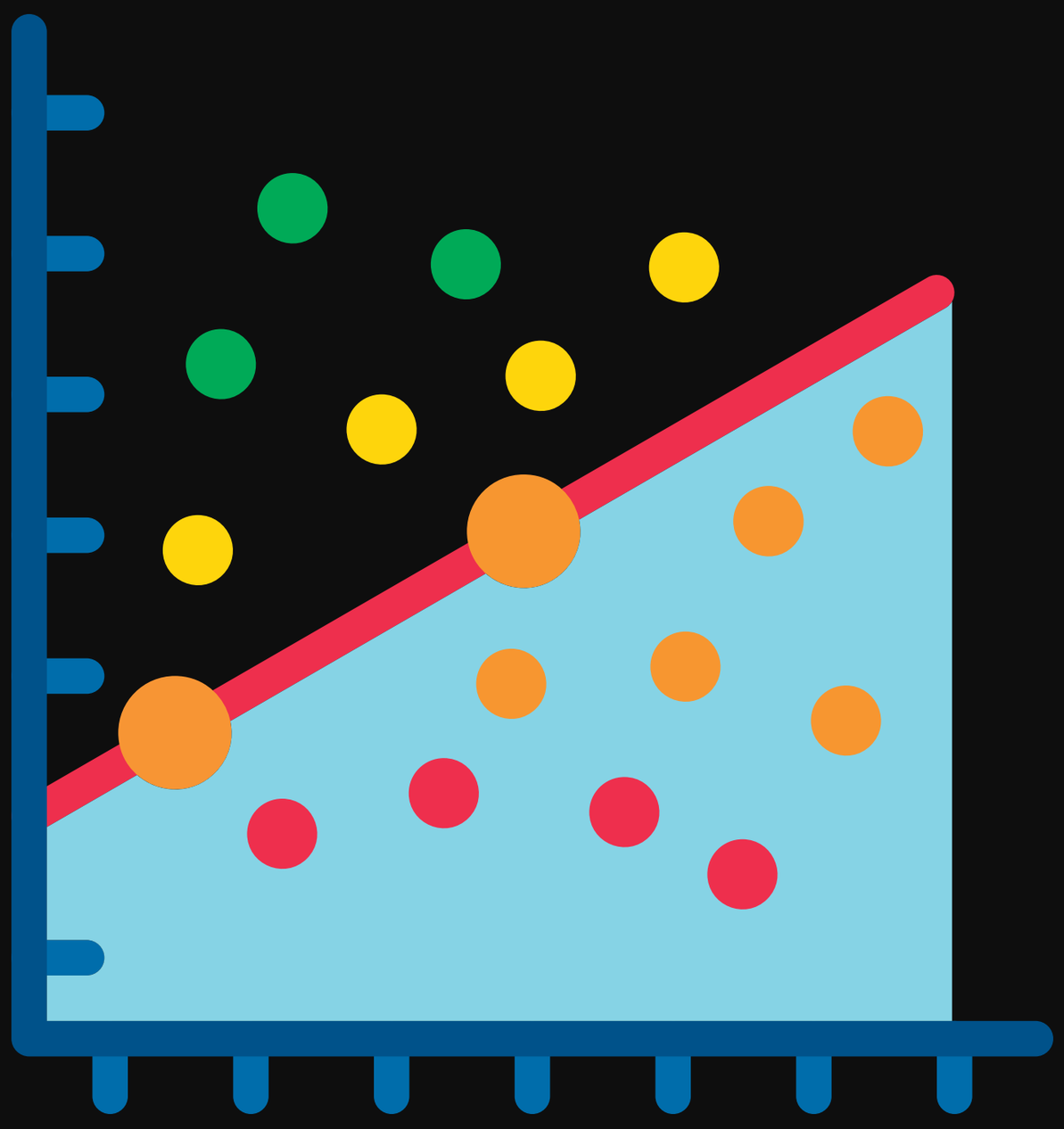
- **Type of data:** Categorical, proportional

- **When to use it:** Use a donut chart to show the proportion of each category.

- **What it shows:** **D**onut charts represent data as slices of a circle, each representing a percentage of the total.

- **When to avoid it:** Avoid using donut charts when there are too many categories or comparing data across groups.

4. Scatterplot



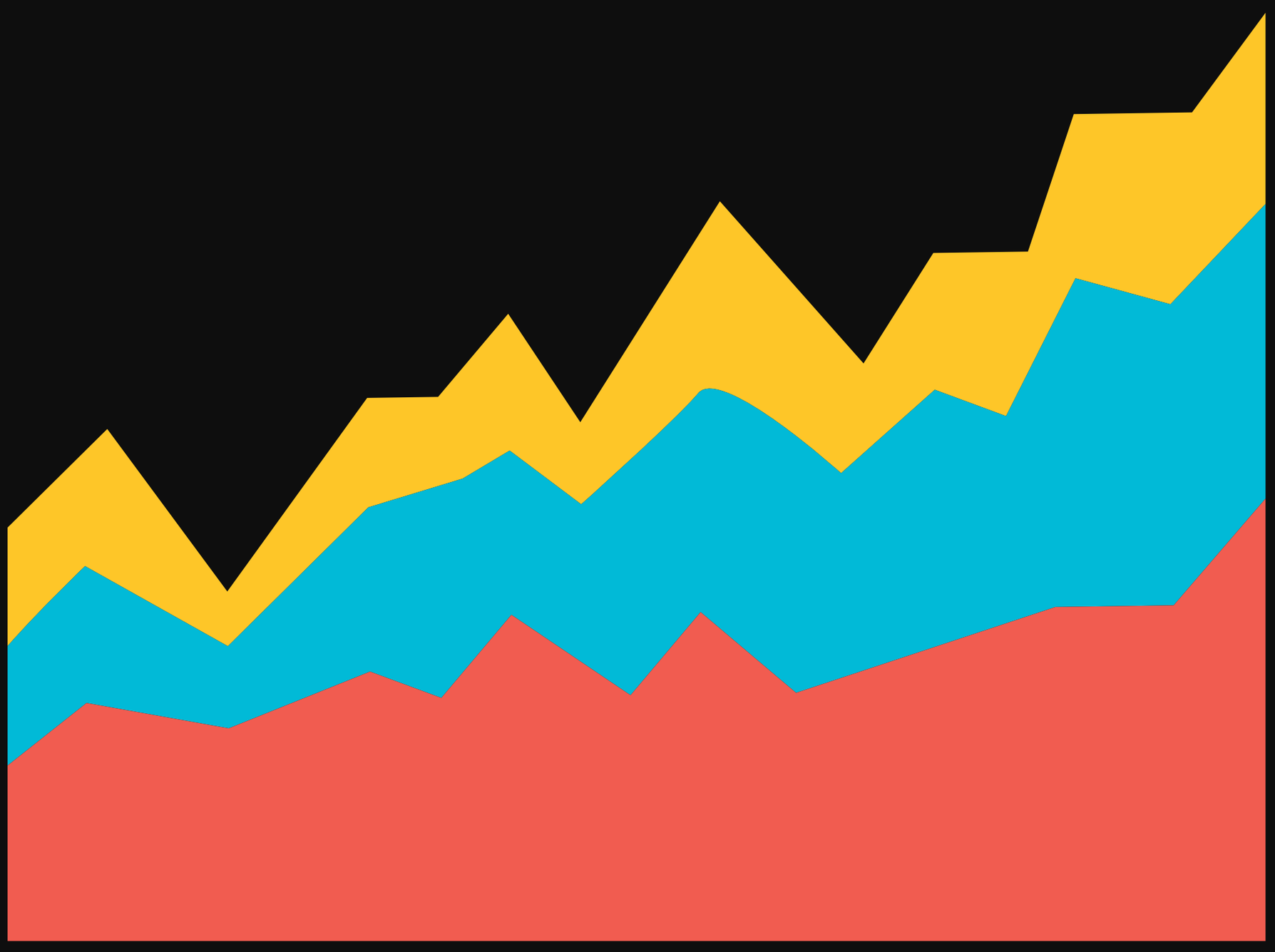
- **Type of data:** Continuous, bivariate

- **When to use it:** Use a scatterplot to display the relationship between two variables.

- **What it shows:** Scatterplots plot data points on a two-dimensional plane, with one variable on the X-axis and the other on the Y-axis.

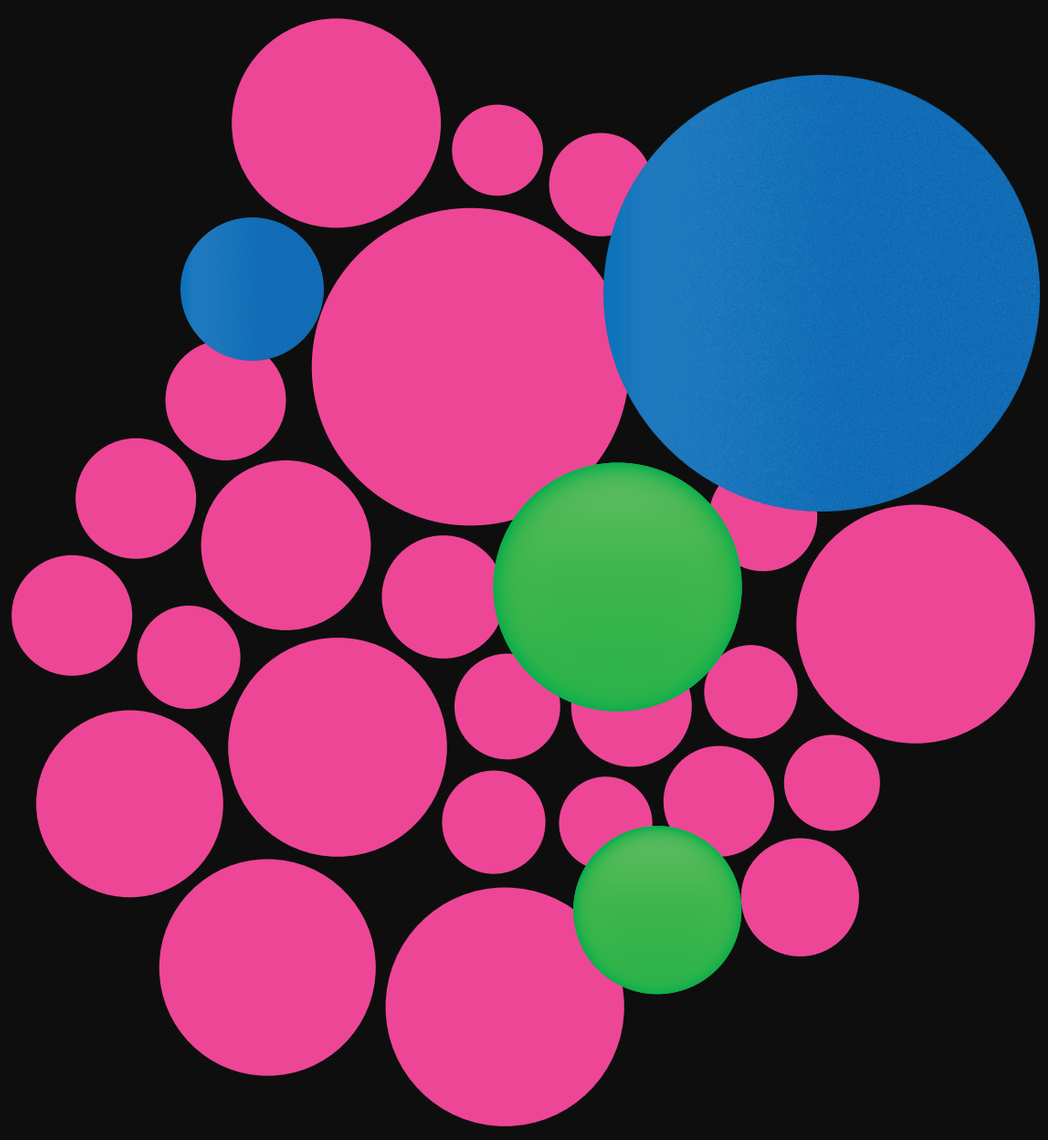
- **When to avoid it:** Don't use a scatterplot when the relationship between variables is irrelevant or when comparing multiple categories.

5. Area Chart



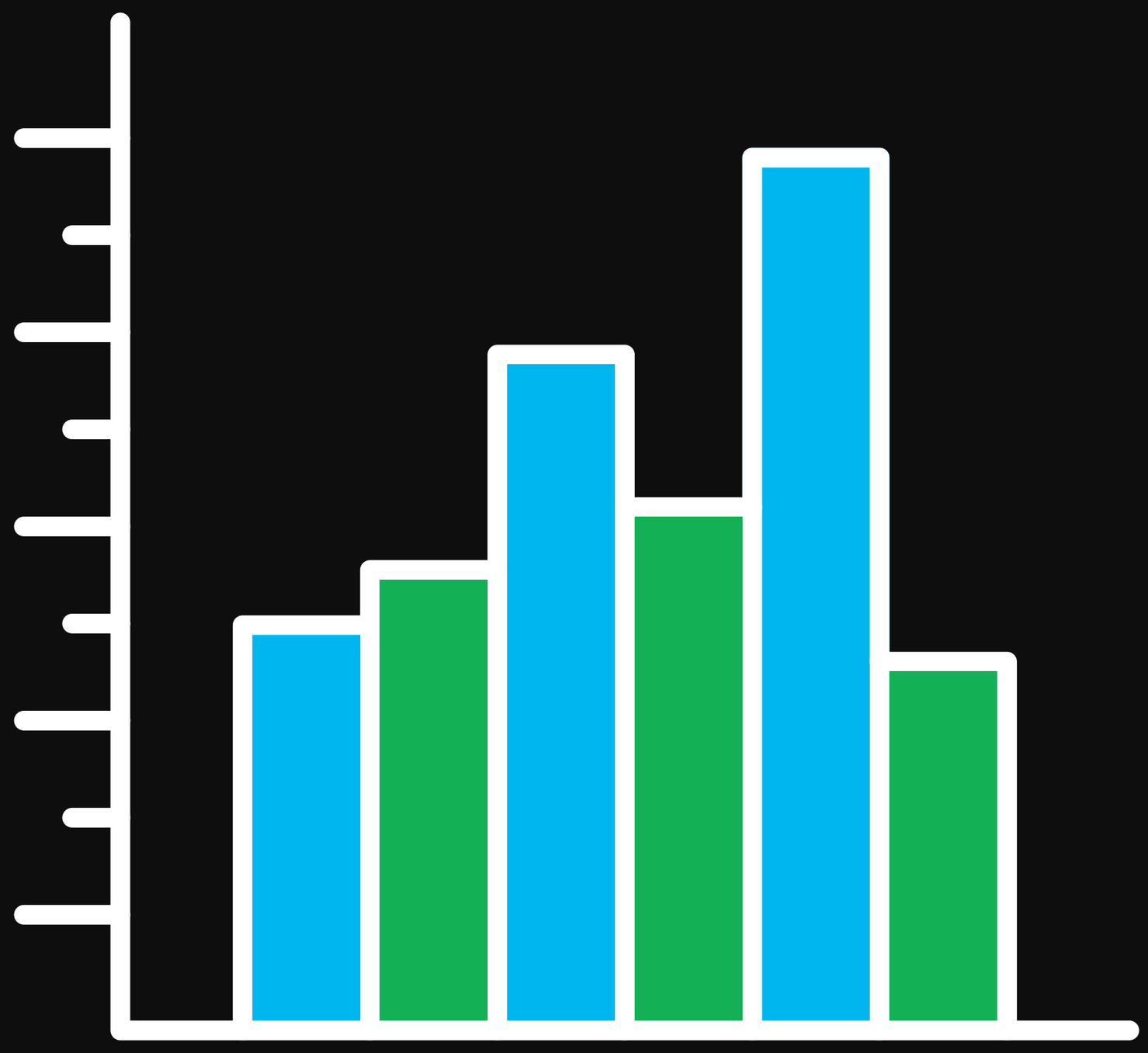
- **Type of data:** Continuous, time-series
- **When to use it:** Use an area chart to show the volume or magnitude of data over time.
- **What it shows:** Area charts are similar to line charts, but the area between the line and the X-axis is filled, emphasizing the volume or magnitude.
- **When to avoid it:** Avoid using an area chart with multiple data series with overlapping areas, as it can be confusing.

6. Bubble Chart



- **Type of data:** Continuous, multivariate
- **When to use it:** Use a bubble chart to display the relationship between three variables.
- **What it shows:** Bubble charts are a variation of scatterplots, with the size of the bubbles representing the third variable.
- **When to avoid it:** Don't use a bubble chart when the size of the bubbles is not meaningful or when comparing multiple categories.

7. Histogram



- **Type of data:** Continuous, univariate

- **When to use it:** Use a histogram to display the data distribution.

- **What it shows:** Histograms are similar to bar charts, but the data is divided into equal intervals, and the bar's height represents the data frequency in each interval.

- **When to avoid it:** Avoid using histograms when the data is categorical or comparing data across groups.

8. Heatmap



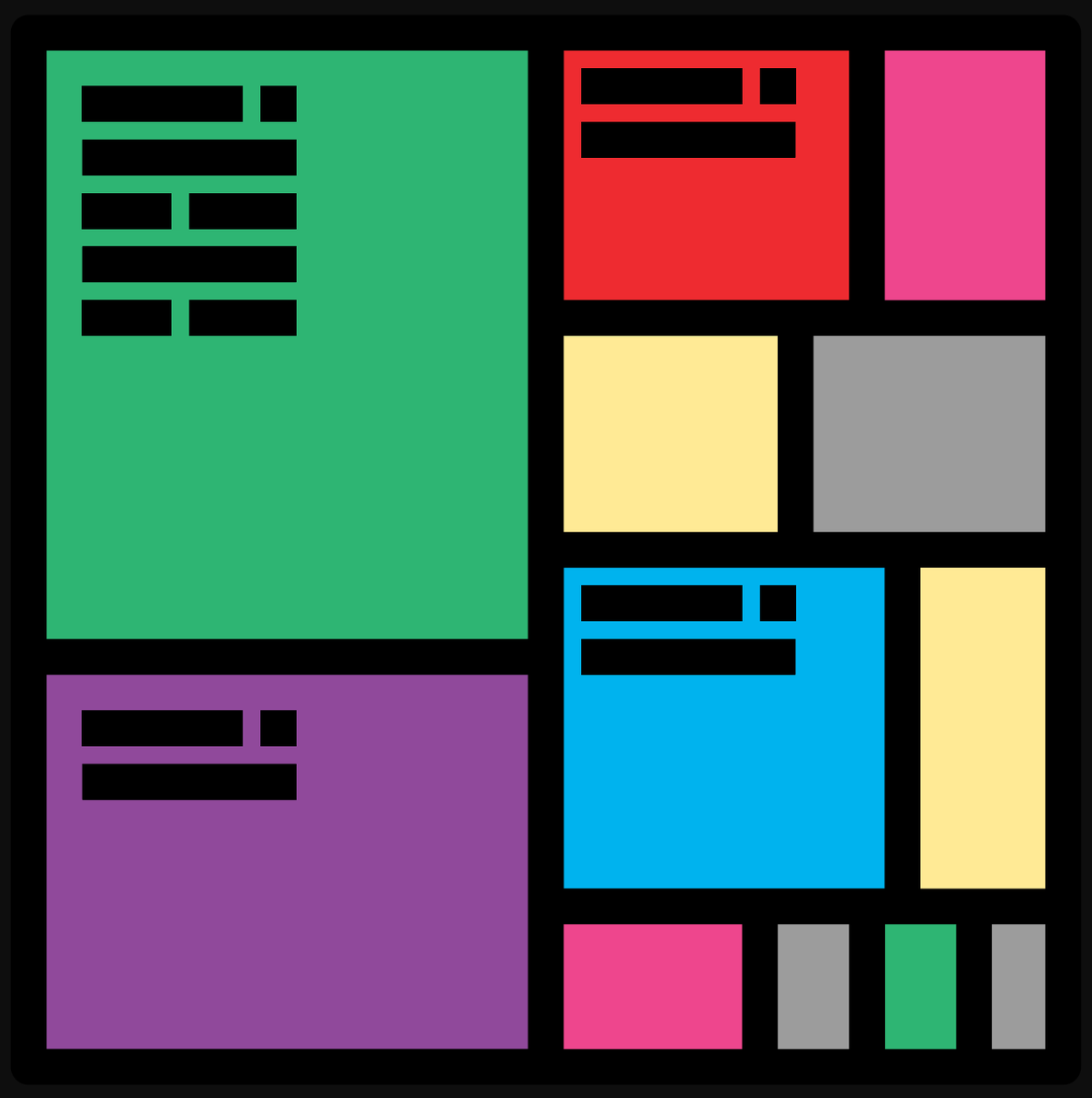
- **Type of data:** Continuous, multivariate

- **When to use it:** Use a heatmap to display the relationship between two variables using color intensity.

- **What it shows:** Heatmaps use a color scale to represent the value of each cell in a matrix, with one variable on the X-axis and the other on the Y-axis. Darker colors indicate higher values, while lighter colors represent lower values.

- **When to avoid it:** Don't use a heatmap when the relationship between variables is irrelevant, when the data is categorical, or when comparing multiple categories.

9. Treemap



- **Type of data:** Categorical, hierarchical

- **When to use it:** Use a treemap to display hierarchical data or to show the proportion of each category as a whole.

- **What it shows:** Treemaps use nested rectangles to represent data, with the size of each rectangle proportional to its value. Color can be used to indicate additional information.

- **When to avoid it:** Avoid using treemaps when there are too many categories or the data is not hierarchical.

10. Radar Chart



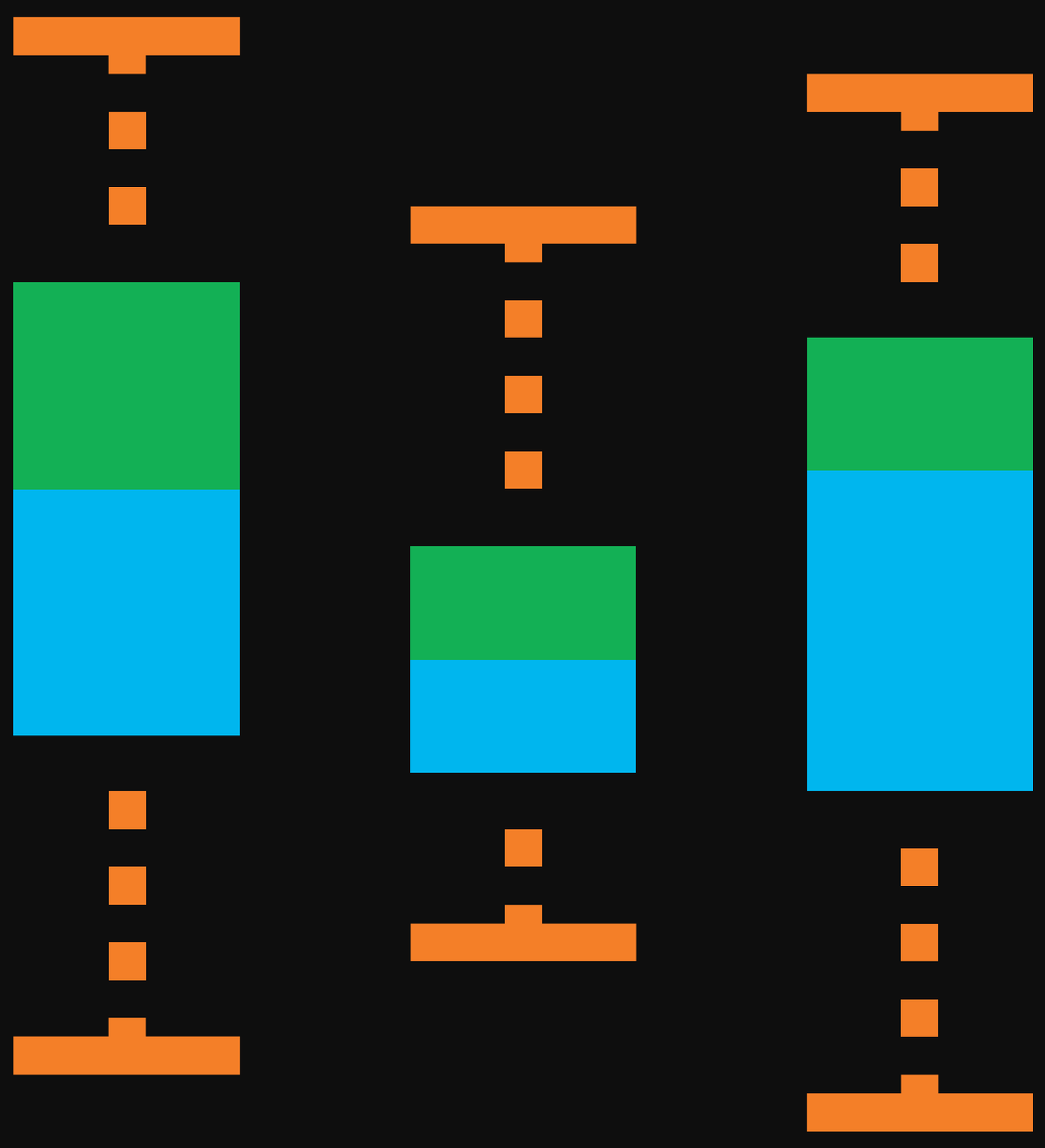
- **Type of data:** Continuous, multivariate

- **When to use it:** Use a radar chart to display the performance or characteristics of different categories across multiple dimensions.

- **What it shows:** Radar charts use a circular layout with multiple axes, each representing a dimension. Data points are plotted on each axis and connected to form a shape.

- **When to avoid it:** Don't use a radar chart when there are only a few dimensions or when comparing data across groups.

11. Box Plot



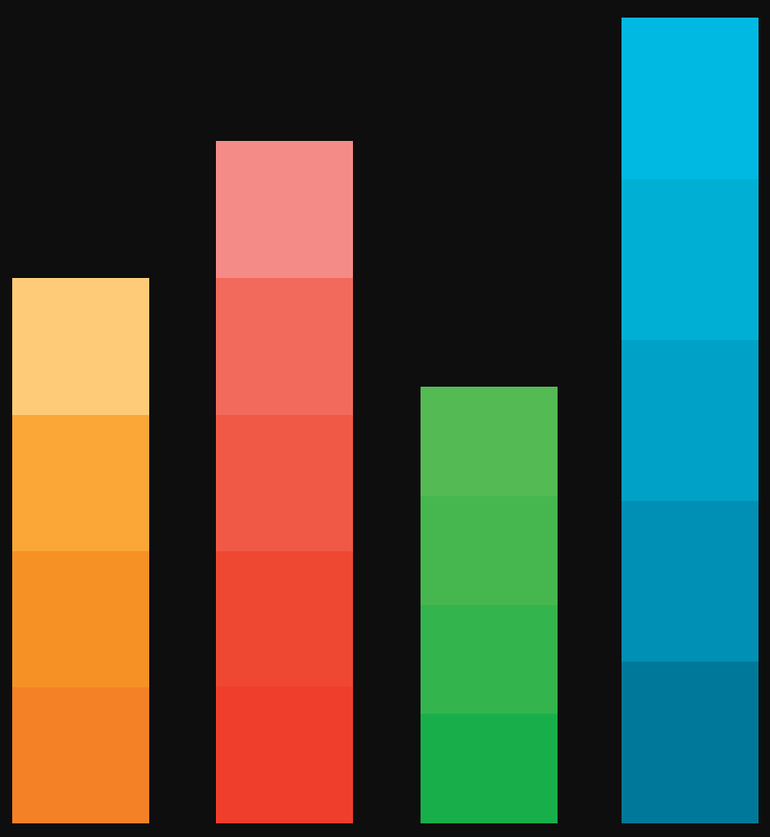
- **Type of data:** Continuous, univariate

- **When to use it:** Use a box plot to display the distribution of data and detect outliers.

- **What it shows:** Box plots use a rectangular box to represent the interquartile range (IQR) and whiskers to show the range of data. The median is represented by a line inside the box, and outliers are plotted as individual points.

- **When to avoid it:** Avoid using box plots when the data is categorical or when showing trends over time.

12. Stacked Bar Chart



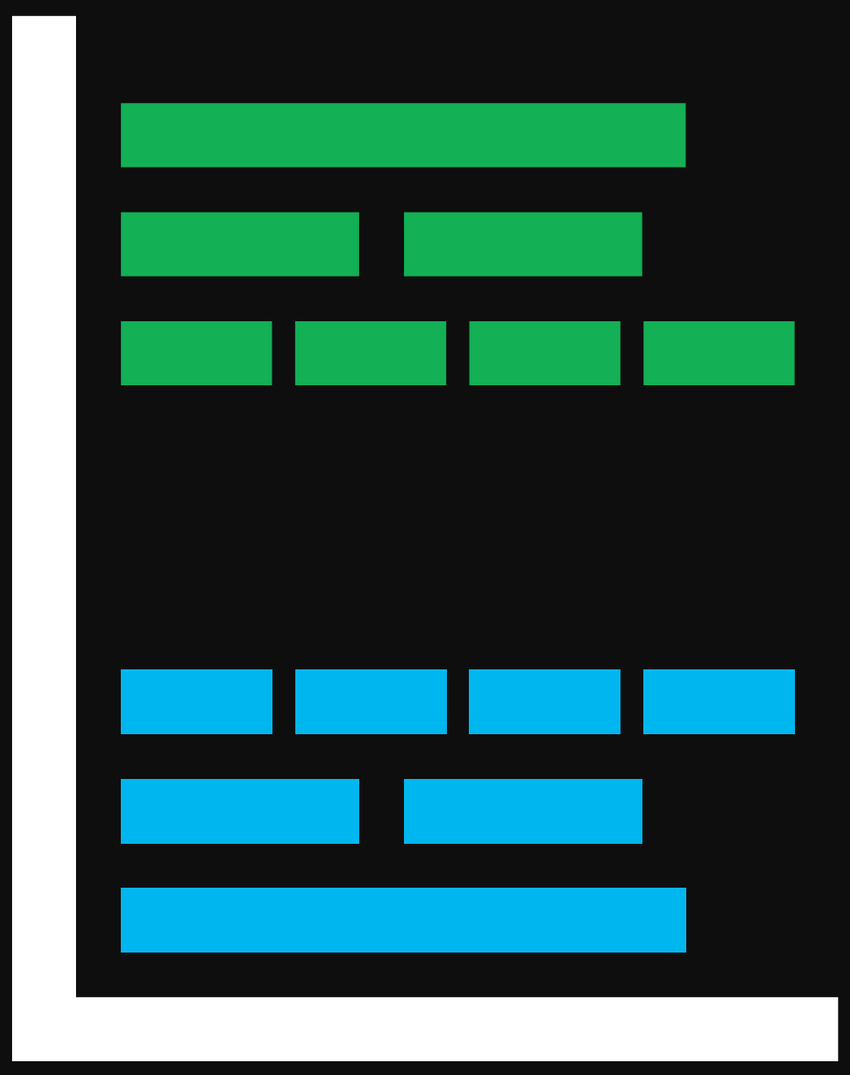
- **Type of data:** Categorical, quantitative

- **When to use it:** Use a stacked bar chart to compare data across categories and show the composition of each category.

- **What it shows:** Stacked bar charts display data using stacked rectangular bars, with the length of each segment representing the value. The total length of the bar represents the sum of all values in a category.

- **When to avoid it:** Don't use a stacked bar chart when there are too many categories or if the data is continuous.

13. Gantt Chart



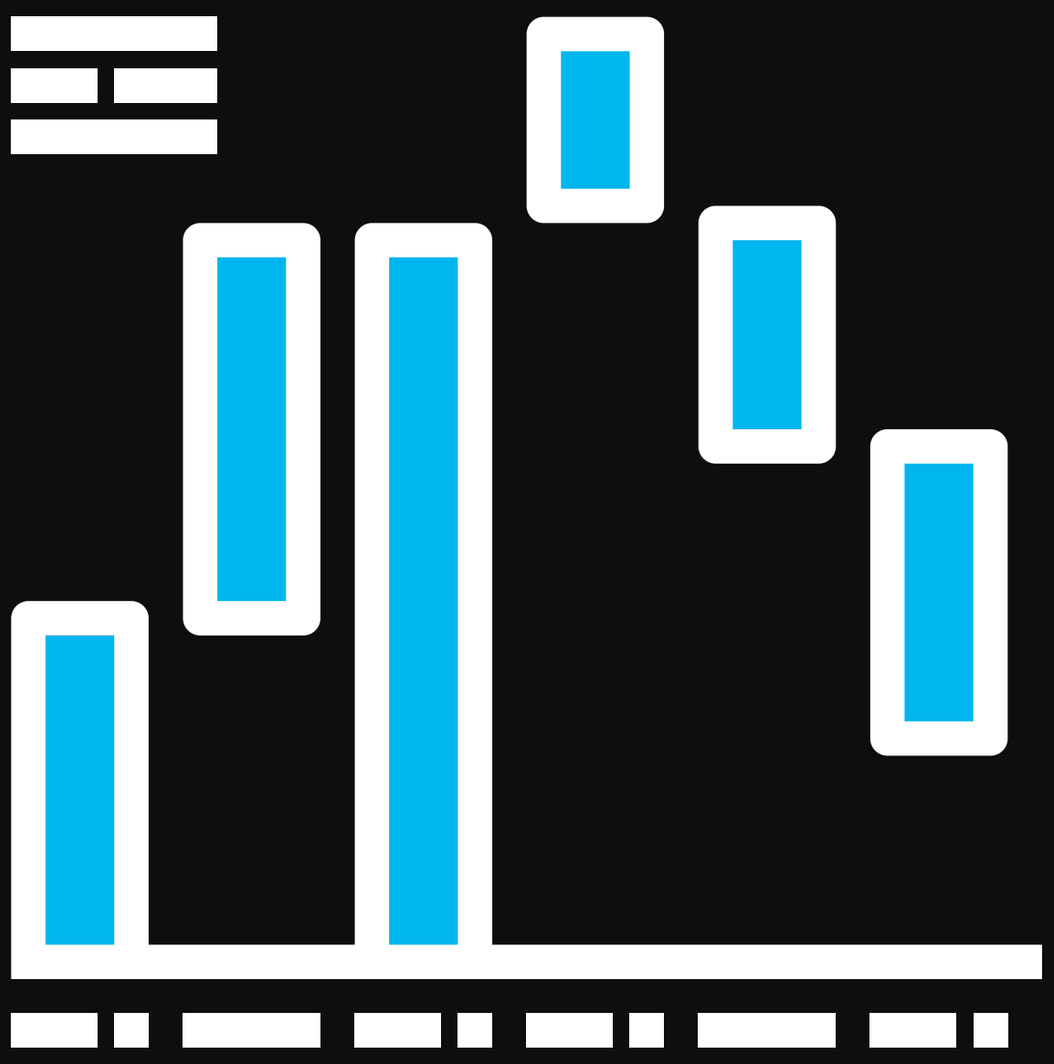
- **Type of data:** Time-based, project management

- **When to use it:** Use a Gantt chart to visualize project schedules, tasks, and milestones.

- **What it shows:** Gantt charts use horizontal bars to represent tasks, with the length of the bar indicating the duration. The X-axis represents time, and tasks can be grouped by category.

- **When to avoid it:** Avoid using Gantt charts when the data is not time-based or when showing relationships between variables.

14. Waterfall Chart



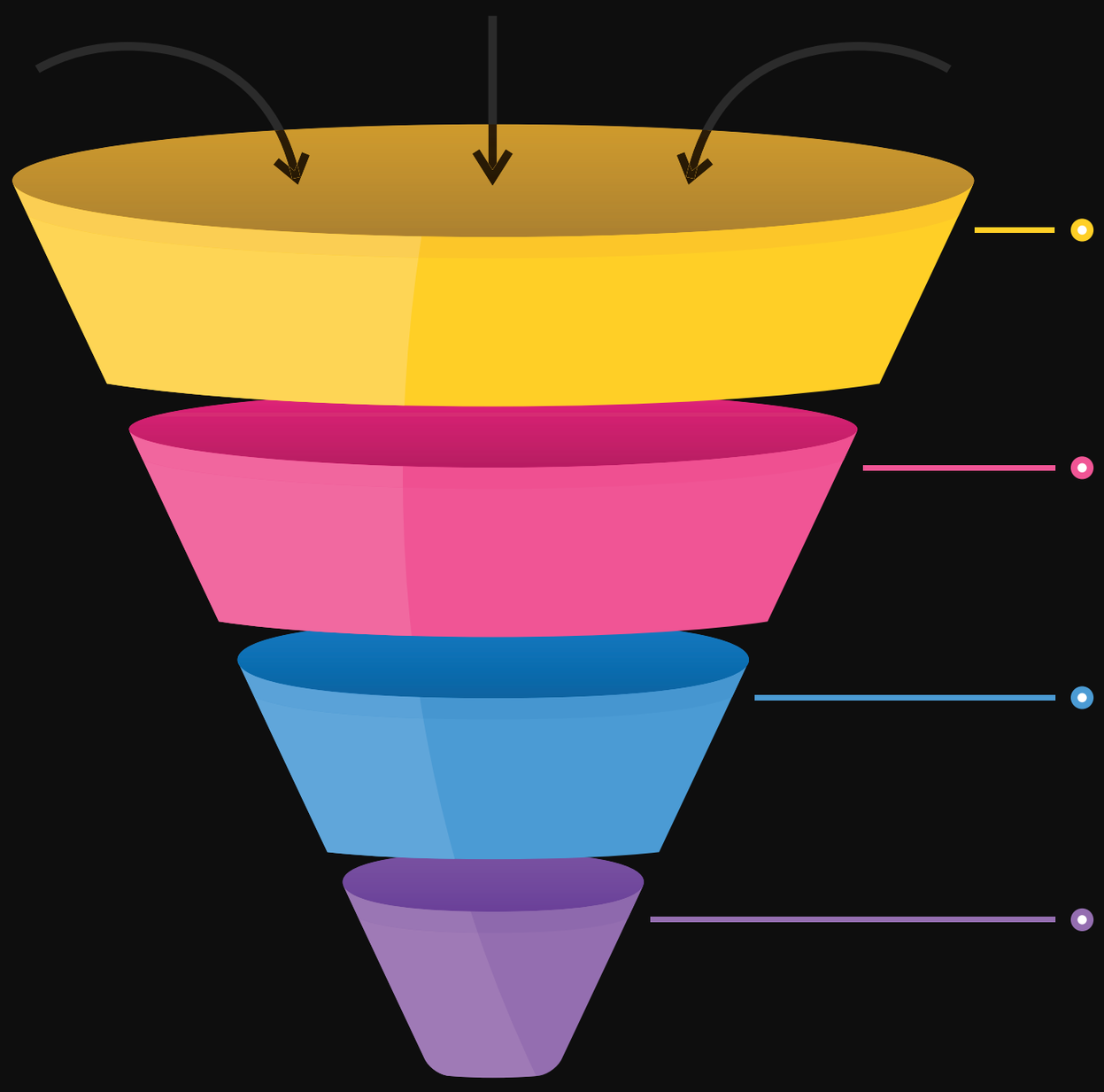
- **Type of data:** Quantitative, sequential

- **When to use it:** Use a waterfall chart to visualize the cumulative effect of sequential data, such as financial or inventory changes.

- **What it shows:** Waterfall charts use vertical bars to represent the value of each step, with the bars connected to show the cumulative effect. Positive and negative values can be represented using different colors.

- **When to avoid it:** Don't use a waterfall chart when there is no logical order or relationship between data points.

15. Funnel Chart



- **Type of data:** Categorical, process stages

- **When to use it:** Use a funnel chart to visualize the stages of a process, such as sales, conversions, or customer journeys.

- **What it shows:** Funnel charts use a series of decreasing trapezoids to represent each stage in a process, with the width of each trapezoid proportional to the number of items at that stage.

- **When to avoid it:** Avoid using funnel charts when there is no straightforward process, or the data is continuous.

Summary: You Did It!

Now that you understand these 15 chart types, you'll be better equipped to choose the correct chart for your data visualization needs.

Remember, selecting the appropriate chart type can make all the difference in how your data is perceived and understood. Keep this guide handy as a reference, and happy visualizing!